

Cloud Computing and Scalable Infrastructure

Introduction

Cloud Computing provides computing services such as storage, networking, and servers over the internet. Major providers include AWS, Microsoft Azure, and Google Cloud Platform.

Cloud Service Models

• IaaS – Infrastructure as a Service • PaaS – Platform as a Service • SaaS – Software as a Service

Real-Time Challenges

1. Downtime and service outages. 2. Security misconfigurations. 3. High cloud cost optimization issues. 4. Scaling applications during peak traffic. 5. Vendor lock-in problems.

Real-Time Examples

• Netflix runs on AWS cloud infrastructure. • Zoom scaled massively during remote work growth. • E-commerce systems auto-scale during Black Friday sales. • Cloud backup systems protect enterprise data.

Cloud Security

Cloud environments require IAM policies, encryption, monitoring, and Zero Trust security architecture.